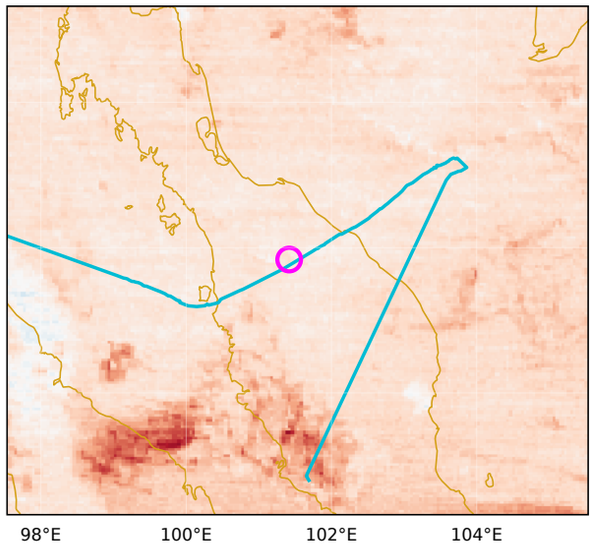
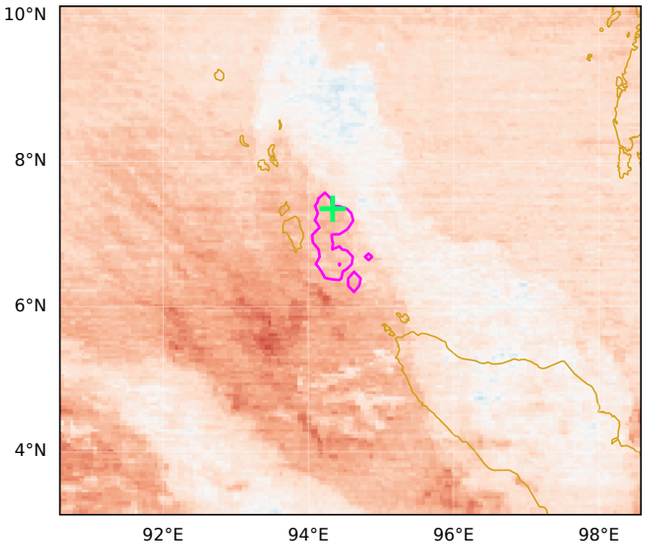
MTSAT-2 split-window difference around the aircraft position at each archived scan, 7-8 March 2014

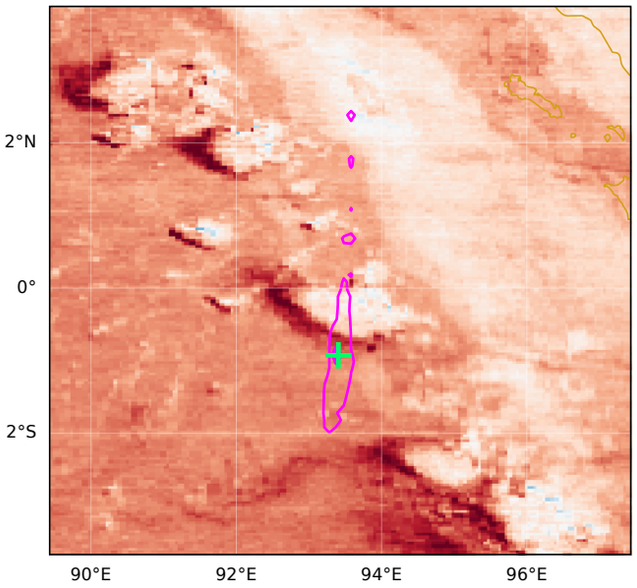
MTSAT-2 full disc, scan start 17:31:15
aircraft reached at 17:43 UTC (median)

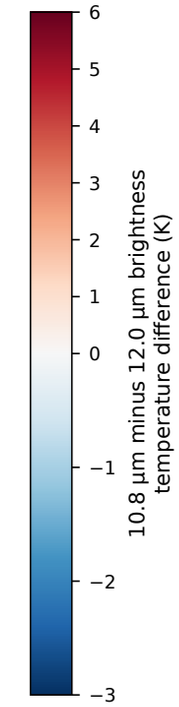


MTSAT-2 full disc, scan start 18:31:15
aircraft reached at 18:42 UTC (median)

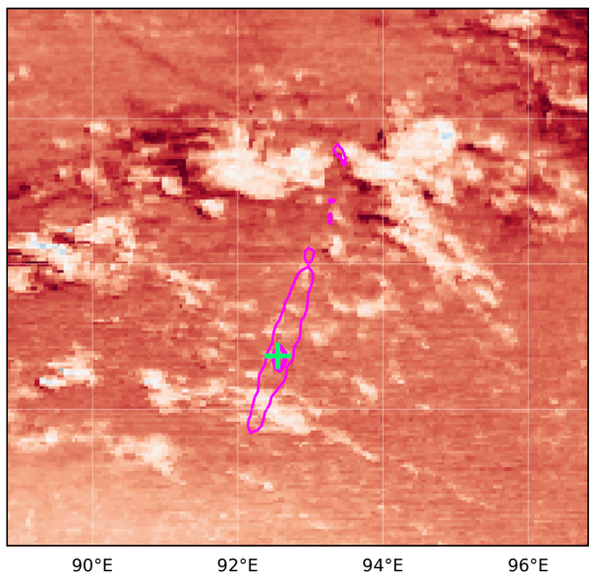


MTSAT-2 full disc, scan start 19:31:15
aircraft reached at 19:44 UTC (median)

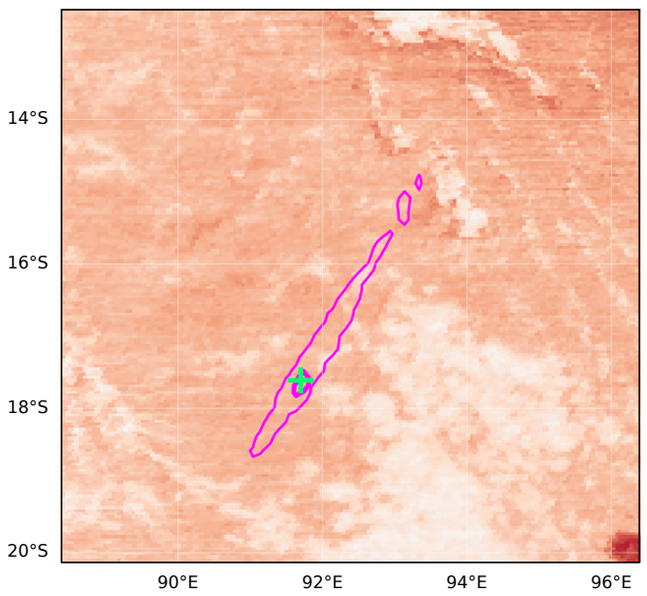




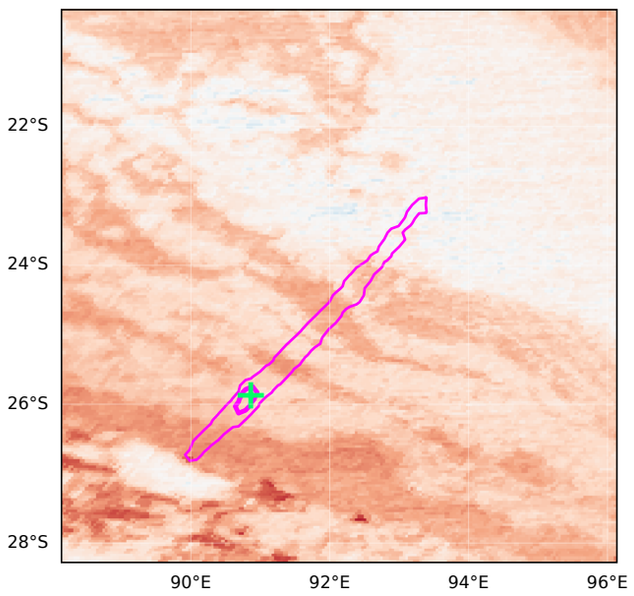
MTSAT-2 full disc, scan start 20:31:15
aircraft reached at 20:46 UTC (median)

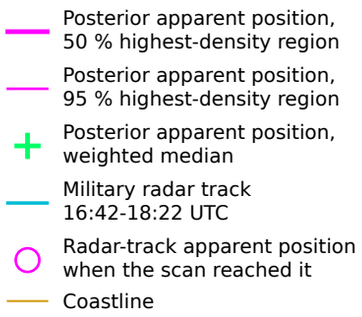


MTSAT-2 full disc, scan start 21:31:15
aircraft reached at 21:48 UTC (median)

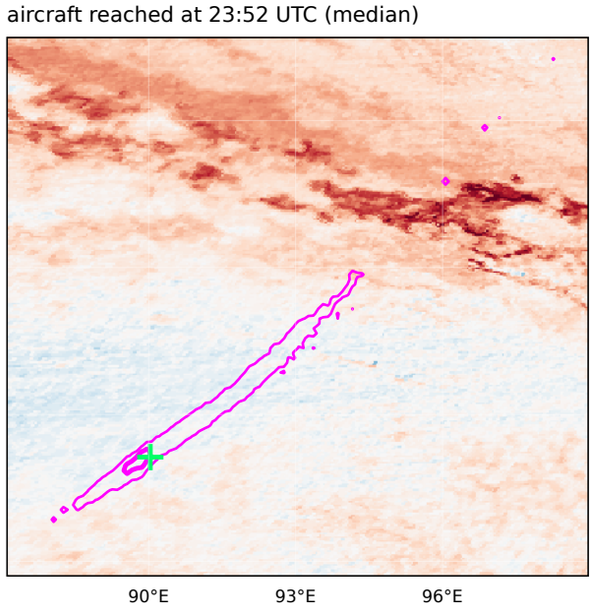


MTSAT-2 full disc, scan start 22:31:15
aircraft reached at 22:50 UTC (median)

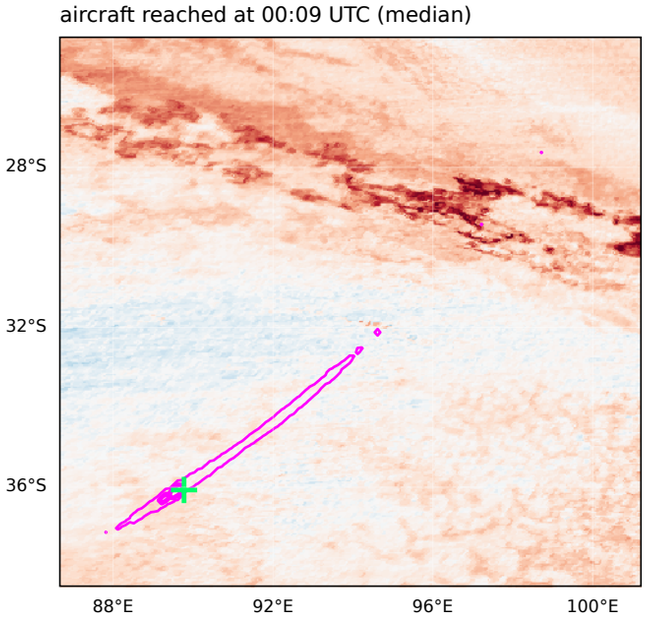




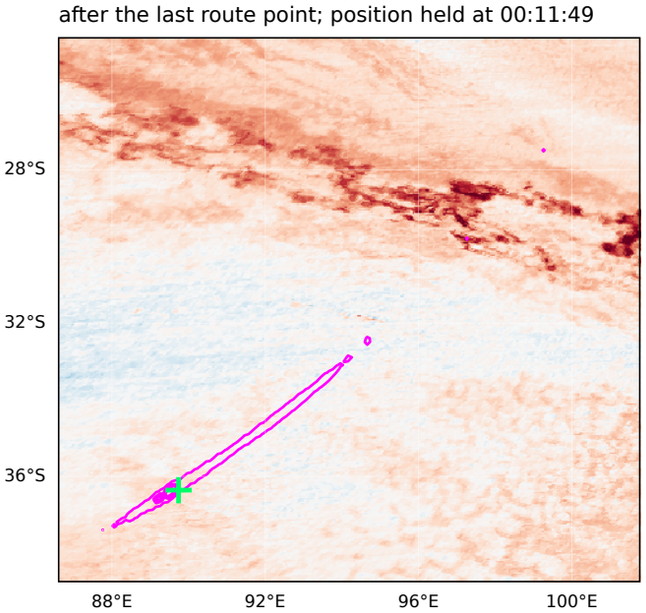
MTSAT-2 full disc, scan start 23:31:15
aircraft reached at 23:52 UTC (median)



MTSAT-2 southern half disc, scan start 00:00:53
aircraft reached at 00:09 UTC (median)



MTSAT-2 southern half disc, scan start 00:14:10
after the last route point; position held at 00:11:49



Plain note: Each panel shows, for one archived image, the difference between two infrared channels. Thin ice cloud and young condensation trails usually show as a stronger red than their surroundings. Clear sea is also red in the moist tropics, so a condensation trail would have to stand out as a thin straight line. The magenta lines show the most probable area for the aircraft. This chart is for inspection; it does not claim a detection.

Technical note: MTSAT-2 (145.0 deg E): Chiba University CEReS gridded V2.0 (0.04 deg, ftp://mtsat.cr.chiba-u.ac.jp/pub/MTSAT-2/gridded_V2.0/), IR1 10.8 um / IR2 12.0 um, calibration table from the JMA HRIT header in each file; full-disc line times from the native HRIT header of the 17:32 scan (N->S, 0.580 s/line), applied to other hours by analogy; southern half-disc start times 00:00:53 and 00:14:10 by analogy with the 18:00:53 header. Aircraft at FL350; 'apparent position' = where the line of sight through the aircraft meets the surface (parallax correction applied so the contours line up with a geolocated image). Posterior: core run J BTO+BFO, 148,000 routes, family weights waypoints 0.9982, free 0.0013, repro-radar 0.0003, descent-climb 0.0002; position at the modelled time the scan line reached each route; routes end 00:11:49 UTC, held there for later scans. Highest-density regions from a 0.1 deg weighted histogram. Before 18:01:49 the radar track (Geoscience Australia) is used. Split-window difference: thin ice cloud and young contrails are usually positive (red). Coastlines: Natural Earth 10 m. Altitude FL300-FL400 would shift the apparent positions by a further 0-15 % of the parallax shown in the sensor table.